

D04. Configuration and Environment Setup Pack

Environment Overview

The project is designed to run as a small, laptop-runnable inbound warehouse commissioning simulation. The working environment is in WSL with Ubuntu. The solution relies on configuration files, Bash scripts, generated runtime outputs, Oracle SQL and PL/SQL artifacts, and a small C++ component. The environment is intentionally simple so that the selected inbound scenario can be configured, executed, and validated without external enterprise systems.

Environment Assumptions

- The project is executed in a Linux-compatible environment.
- Bash is available for running the operational scripts.
- The C++ compiler MSYS2 is available if the C++ utility needs to be compiled.
- The Oracle SQL and PL/SQL parts are delivered as scripts and design artifacts for the selected flow.
- The project is run locally on team laptops using free and publicly available tools.

Main Folder Structure

The setup of the project uses the following logical folder structure:

- **config/:** configuration and parameter files
- **scripts/:** setup, run, validation, and reset scripts
- **runtime/:** generated logs, snapshots, and output files
- **sql/:** database schema, sample data, and inspection queries
- **plsql/:** selected procedural database logic
- **cpp/:** small focused C++ utility
- **docs/:** project documentation and deliverables

Basic Connectivity and Environment Checks

Before running the full demo, these checks should be performed:

- Confirm that Linux WSL2 is available
- Confirm that the Bash scripts are able to be executed
- Confirm that the required configuration files are readable

- Confirm that runtime folders are created
- Confirm that the C++ utility can be compiled and executed
- Confirm that the database (SQL) scripts can be compiled and executed

Run Instructions

The project is expected to use a small set of Bash scripts for normal execution:

- **setup_demo.sh** prepares the environment and runtime folders
- **run_demo.sh** executes the selected inbound scenario
- **validate_demo.sh** checks whether the final state matches the expected result
- **reset_demo.sh** clears generated outputs so that the demo can be repeated

The execution sequence should be:

1. run the setup script
2. run the main demo script
3. run the validation script
4. reset the environment if another clean run is needed

Expected Outputs

- Status logs
- Pallet state snapshots
- Location usage snapshots
- Validation result files or pass/fail messages